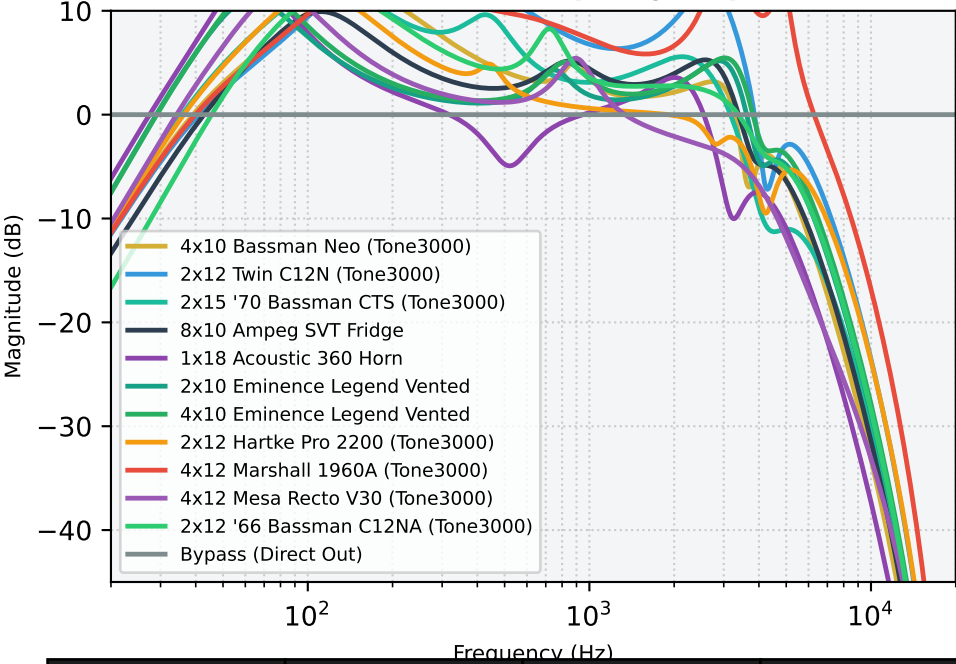


BOSSMAN — ACOUSTIC CABINET MODELS BENCHMARK

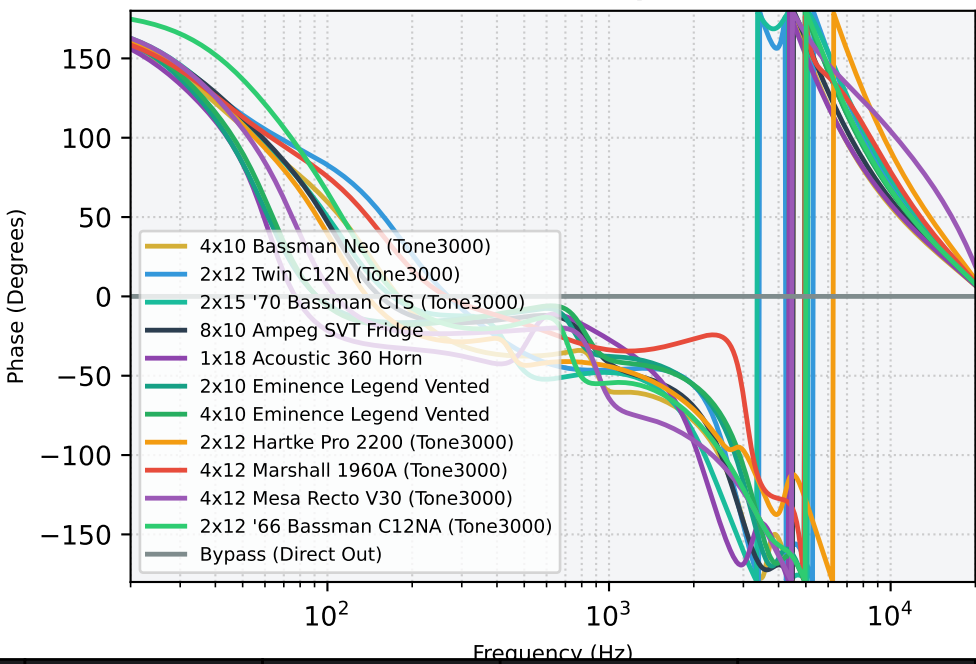
Frequency & Phase Responses of 5 Modeled Enclosures + Direct Bypass (Pure Biquad Bank)

Zero-ML / Zero-Latency

Cabinet Enclosure Frequency Response



Cabinet Phase Responses



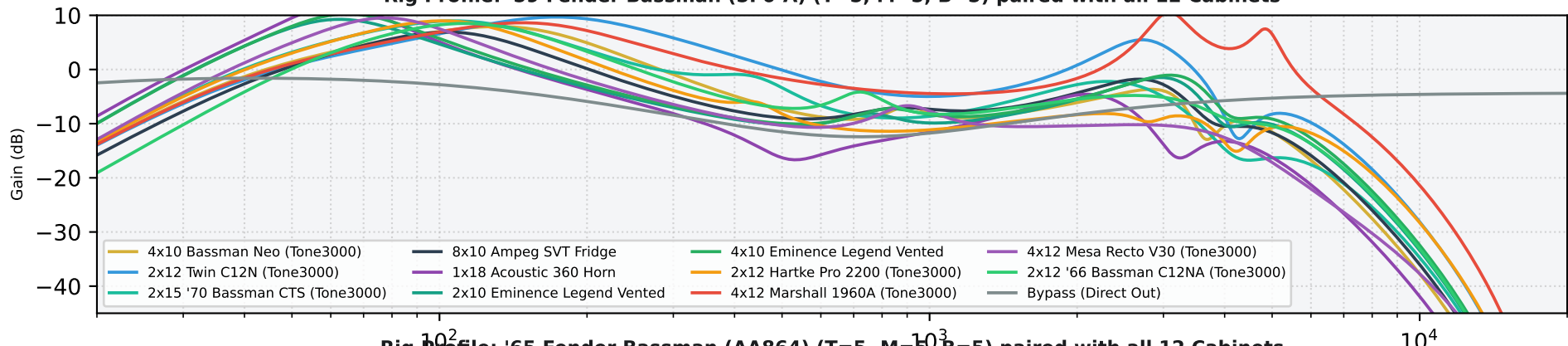
Cabinet Model	Enclosure Type	Cone Thump Freq	Thump Boost	Mid Contour	Presence Peak	Inductance Rolloff	Sonic Role
4x10 Bassman Neo	Open-back Pine	149.5 Hz	+12.0 dB	900 Hz (+3.5 dB)	3373 Hz (+7.7 dB)	4000 Hz	Punchy fast transient bass articulation
2x12 Twin C12N	Open-back Birch	180.0 Hz	+12.0 dB	443 Hz (+6.0 dB)	2750 Hz (+12.4 dB)	5139 Hz	Deep Blackface lows, singing glassy bell
2x15 '70 Bassman CTS	Deep Sealed Pine	116.0 Hz	+12.0 dB	440 Hz (+6.0 dB)	2423 Hz (+7.2 dB)	4408 Hz	Sub-bass authority, massive physical chest kick
8x10 Ampeg SVT Fridge	Infinite Baffle 8x10	110.0 Hz	+10.0 dB	850 Hz (+4.2 dB)	2850 Hz (+6.8 dB)	4290 Hz	Sub-low-mid punch, tight unported transient sl
1x18 Acoustic 360 Horn	Folded Rear-Horn	62.0 Hz	+14.0 dB	520 Hz (-5.5 dB)	2150 Hz (+5.0 dB)	3600 Hz	Massive Jaco sub-bass acoustic compression thro
2x10 Eminence Legend Vented	Vented Box 3 cu.ft	62.0 Hz	+11.5 dB	750 Hz (+3.8 dB)	3200 Hz (+7.5 dB)	4500 Hz	Lucas vented design, deep 62Hz Helmholtz t
4x10 Eminence Legend Vented	Vented Box 6 cu.ft	65.0 Hz	+12.5 dB	820 Hz (+4.5 dB)	3400 Hz (+8.2 dB)	4600 Hz	Lucas vented design, massive low punch 65
2x12 Hartke Pro 2200	Ported Dual-Chamber	108.0 Hz	+12.0 dB	447 Hz (+2.5 dB)	2797 Hz (-2.3 dB)	5210 Hz	Minimum cone lightning slap transient & punch
4x12 Marshall 1960A	Closed-back 1960	161.0 Hz	+12.0 dB	689 Hz (+6.0 dB)	3073 Hz (+15.0 dB)	6153 Hz	Creamy British roar, forward aggressive bite
4x12 Mesa Recto V30	Closed Oversized	78.4 Hz	+12.0 dB	900 Hz (+6.0 dB)	3800 Hz (+3.9 dB)	4915 Hz	Tight modern percussive low thump & scooped bi
2x12 '66 Bassman C12NA	Closed Vintage 66	121.3 Hz	+12.0 dB	724 Hz (+6.0 dB)	2632 Hz (+2.7 dB)	4520 Hz	Vintage acoustic resonance, velvety midran
Bypass	Direct Line	Flat	0.0 dB	Flat	Flat	20.0 kHz	For external IR loader plugins

IRONSTACK — END-TO-END RIG MATRIX (PART 1: 12 CAB COMBINATIONS PER HEAD)

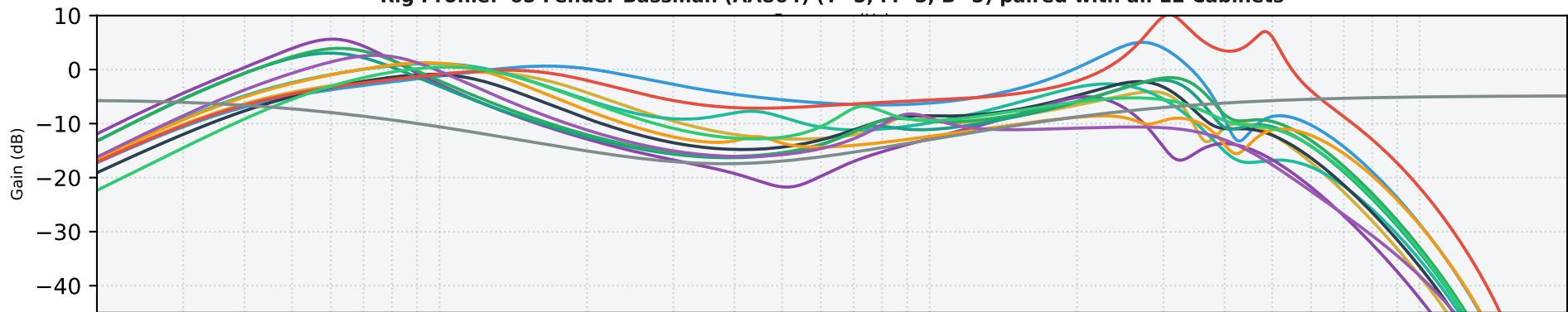
Combined Tone Stack + Speaker Cabinet Responses across 12 Acoustically Modeled Enclosures

Status: **VALIDATED** (120 Rigs)

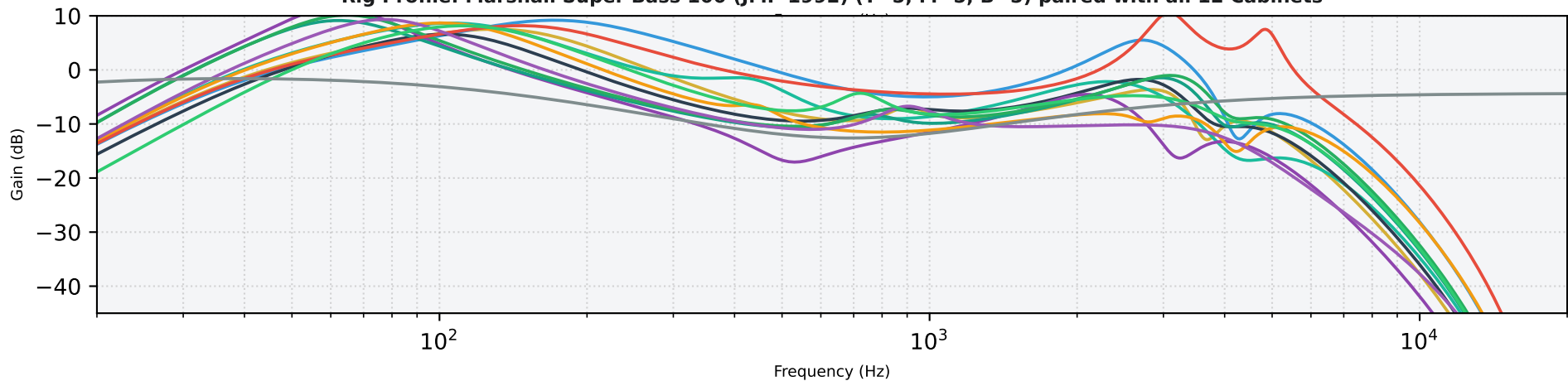
Rig Profile: '59 Fender Bassman (5F6-A) (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: '65 Fender Bassman (AA864) (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Marshall Super Bass 100 (JMP 1992) (T=5, M=5, B=5) paired with all 12 Cabinets

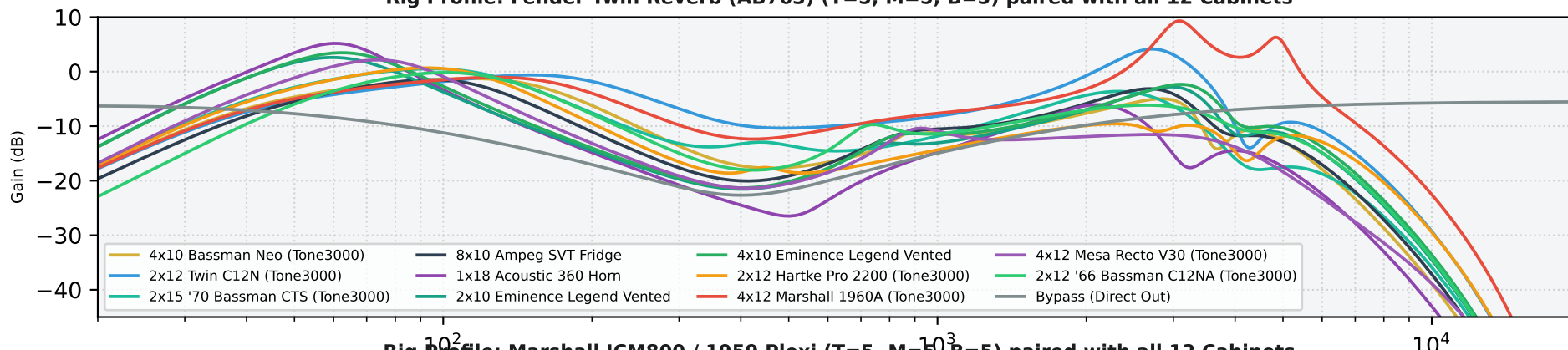


IRONSTACK — END-TO-END RIG MATRIX (PART 2: 12 CAB COMBINATIONS PER HEAD)

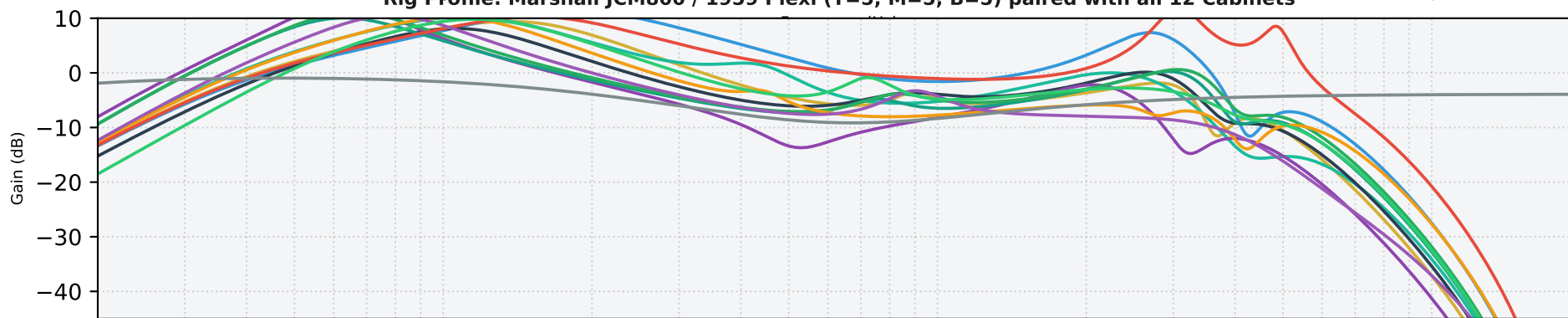
Combined Tone Stack + Speaker Cabinet Responses across 12 Acoustically Modeled Enclosures

Status: **VALIDATED** (120 Rigs)

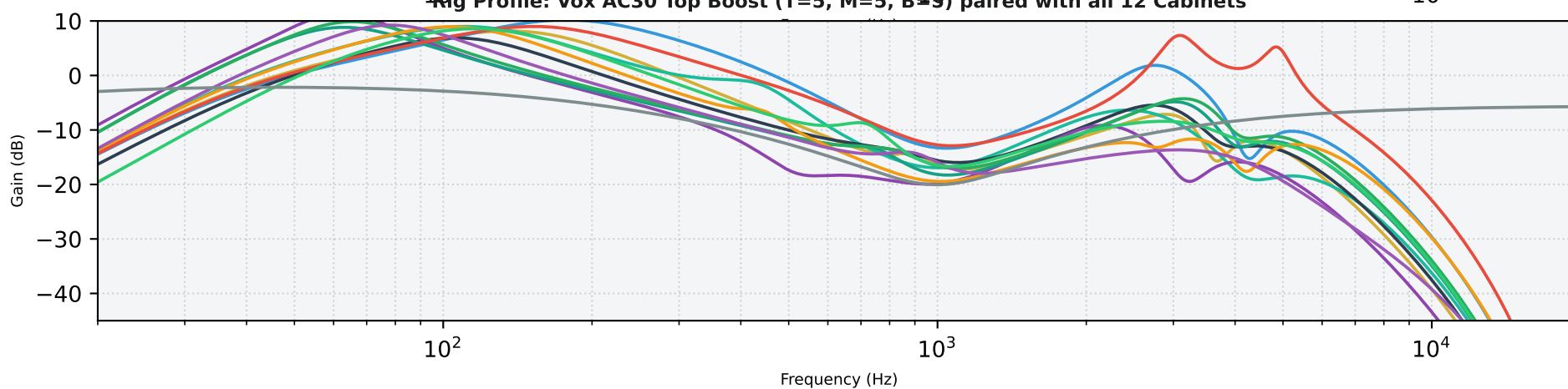
Rig Profile: Fender Twin Reverb (AB763) (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Marshall JCM800 / 1959 Plexi (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Vox AC30 Top Boost (T=5, M=5, B=5) paired with all 12 Cabinets

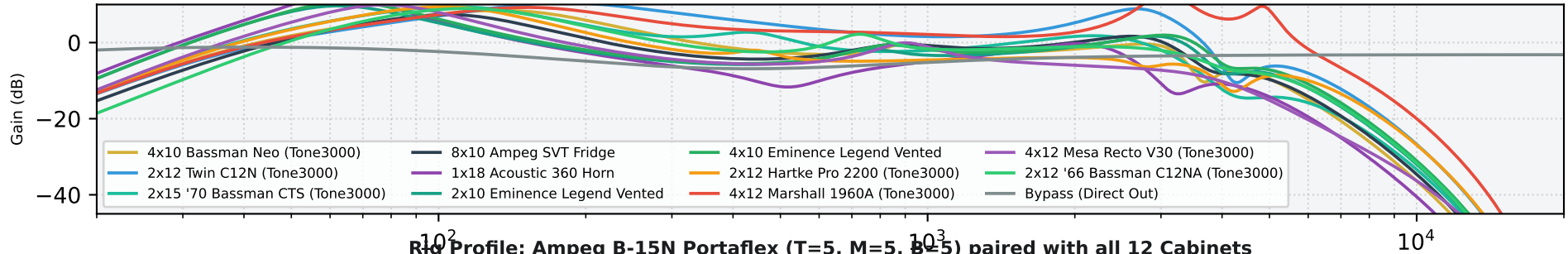


IRONSTACK — END-TO-END RIG MATRIX (PART 3: 12 CAB COMBINATIONS PER HEAD)

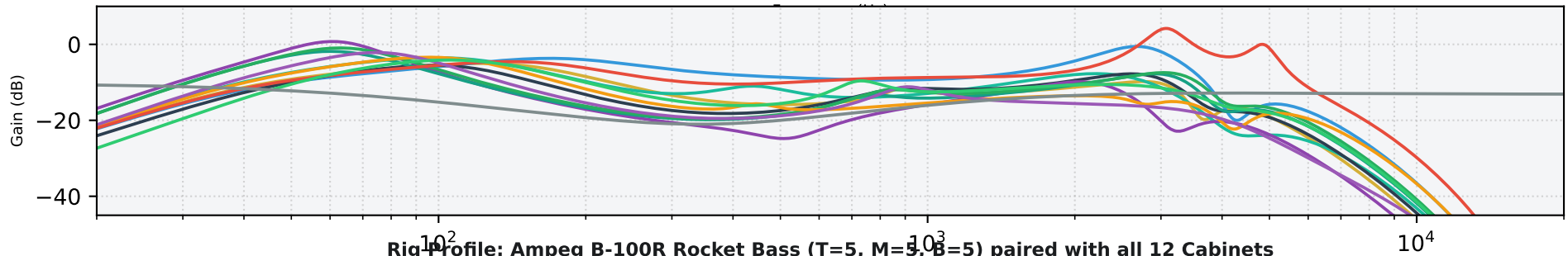
Combined Tone Stack + Speaker Cabinet Responses across 12 Acoustically Modeled Enclosures

Status: **VALIDATED** (120 Rigs)

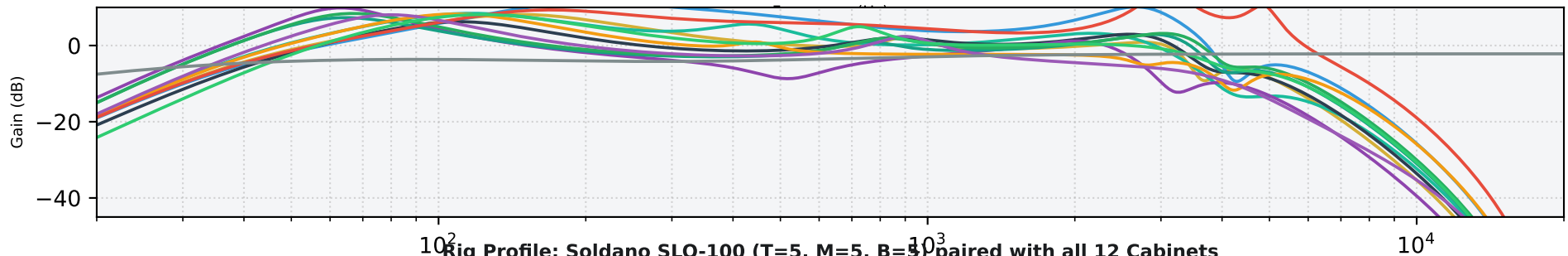
Rig Profile: Mesa Boogie Dual Rectifier (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Ampeg B-15N Portaflex (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Ampeg B-100R Rocket Bass (T=5, M=5, B=5) paired with all 12 Cabinets



Rig Profile: Soldano SLO-100 (T=5, M=5, B=5) paired with all 12 Cabinets

